$$x + \Delta x = \sin(y + \Delta y).$$

$$\text{Therefore, } \frac{\Delta y}{\Delta x} = \frac{\Delta y}{\sin(y+\Delta y) - \sin y} = \frac{1}{\frac{\sin(y+\Delta y)-\sin y}{\Delta y}}$$

As $\Delta x \to 0$, $\Delta y \to 0$ also, so that

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

$$= \lim_{\Delta y \to 0} \frac{1}{\frac{\sin(y+\Delta y)-\sin y}{\Delta y}}$$

$$= \frac{1}{\cos y}$$

$$= \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

That is, $\boxed{\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}}$.

### (ii) The derivative of arccos $x$ or $\cos^{-1} x$

We know the identity :

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}.$$

$$\text{This implies, } \frac{d}{dx}(\sin^{-1} x + \cos^{-1} x) = \frac{d}{dx}\left(\frac{\pi}{2}\right) = 0$$

$$\text{This implies, } \frac{d}{dx}(\sin^{-1} x) + \frac{d}{dx}(\cos^{-1} x) = 0$$

$$\text{Therefore, } \frac{1}{\sqrt{1-x^2}} + \frac{d}{dx}(\cos^{-1} x) = 0$$

or $\boxed{\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}}$.

### (iii) The derivative of arctan $x$ or $\tan^{-1} x$

$$\text{Let } y = f(x) = \tan^{-1} x . \qquad \ldots (1)$$

$$\text{This implies, } y + \Delta y = f(x + \Delta x) = \tan^{-1}(x + \Delta x) \qquad \ldots (2)$$

$$x = \tan y \text{ and}$$

$$x + \Delta x = \tan(y + \Delta y)$$



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